

Anticipating Organic Search Decay: A Time-Aware Machine Learning Framework for Proactive Content Refresh Prioritization

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Abstract

When organic search traffic starts dropping, most content teams only notice weeks or months after the damage is done. This paper asks a straightforward question: can we spot that decline before it happens? Working with the FlyRank Search Intelligence warehouse—a dataset of over 78.8 million daily performance records spanning 18 months—we built 23 time-lagged features from search impressions, rankings, click-through rates, and engagement signals, then trained a binary classifier under a strict chronological validation protocol so that no future data could leak into training. Our balanced logistic regression model, tested on 57,458 completely held-out content-month observations, achieved 49.67% precision, 29.27% recall, an F1 of 0.3684, and an ROC-AUC of 0.5409—clearly beating a naïve baseline that flags nothing at all. We then turned these predictions into a practical three-tier action engine with human-readable reason codes, giving marketing teams a clear, prioritized list of pages to review before traffic actually falls off.

1 Introduction and Problem Statement

Search engine optimization has long been a reactive game [1]. In practice, digital publishers and content marketing teams keep an eye on their analytics dashboards and only realize traffic has dropped weeks or even months after search engines started ranking their pages lower. Researchers have known for years that user behavior signals—click patterns, session data, engagement metrics—carry genuine predictive power for page popularity [2, 3]. Yet putting these signals to work to actually forecast content decay before it happens remains a largely unsolved problem. And the cost of being late is real: once a page loses its search rankings, clawing back to the top takes extensive rewrites, new backlinks, and months of waiting for re-indexing.

A better approach would be to predict which pages are heading for trouble and act before the decline sets

in. Instead of auditing content portfolios on a hunch or scrambling after a traffic drop shows up in last month’s report, editorial teams could direct their effort toward the pages that need it most—right when it matters.

The core operational question driving this work is simple: *Which content items in a publisher’s catalog should be flagged for editorial review in the coming month?* More precisely, we ask:

Can we use historical search visibility and on-page engagement data to identify content that will lose 20% or more of its organic sessions next month—before that drop actually occurs?

By framing content decline as a supervised classification problem, this study makes two main contributions:

1. **A Leakage-Audited, Time-Aware Modeling Pipeline:** We designed a reproducible data engineering workflow using DuckDB on a multi-tenant enterprise data warehouse. Every split is strictly chronological, so the model never sees the future during training.
2. **A Decision-Support Content Refresh Engine:** We built an interpretable scoring system that converts raw decline probabilities into editorial playbooks—things like “rewrite your meta descriptions” or “run a technical audit”—with clear, auditable reason codes attached to every recommendation.

One thing worth being upfront about: this work does not try to reverse-engineer how Google ranks pages, and it does not claim that refreshing content will guarantee traffic recovery. What it does is provide an empirical decision-support tool based on patterns we observe in the data.

2 Data and Public Safety

2.1 Data Source and Scale

All experiments in this study draw on the FlyRank ML Internship Data Warehouse release [4], which is publicly

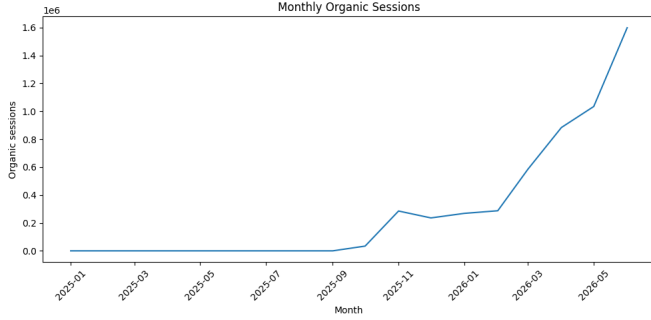


Figure 1: Total organic search sessions across all tenants by calendar month (Jan 2025 – Jun 2026).

hosted on Hugging Face. The main table we work with is `fact_content_daily_performance`, a partitioned Parquet dataset that covers 18 consecutive months from January 2025 through June 2026.

The raw table contains **78,835,655** rows in total. Each row records a single day’s performance for one anonymized content entity, combining Google Search Console (GSC) visibility data with Google Analytics 4 (GA4) traffic and engagement metrics.

2.2 Aggregation and Unit of Analysis

Content strategy decisions happen on a monthly cycle, not day by day. So we aggregated the daily records up to the **content-month** level. For each content entity i in month m , we computed summary statistics using DuckDB [5]:

$$\text{Metrics}_{i,m} = \left\langle \sum \text{GSC}, \sum \text{GA4}, \text{Avg}(\text{Position}) \right\rangle \quad (1)$$

This step reduced 78.8 million daily records down to **2,871,202** content-month rows covering **409,205** unique content entities. Figure 1 shows how total monthly organic traffic grew over the observation window—starting below 10,000 sessions in early 2025 and climbing past 1.59 million by June 2026 as more enterprise clients came onboard.

2.3 Inclusion Filters and Cohort Construction

Not every content entity belongs in a predictive model. To build a reliable cohort for supervised learning, we applied three sensible filters:

1. **Prior-Month Presence:** The content must have had active performance in the previous month ($t-1$), so that lagged predictor signals actually exist.
2. **Minimum Visibility:** Very low-traffic pages introduce too much noise—dropping from 2 clicks to 0 is technically a 100% drop, but it is not meaningful. We required at least 100 search impressions and 10 sessions in the prior month.

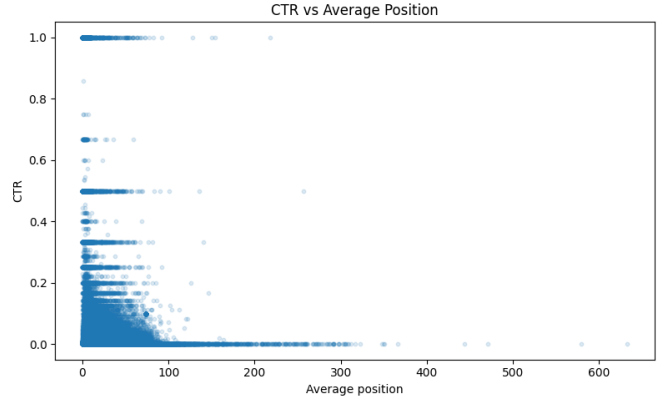


Figure 2: How click-through rate relates to average SERP position across the observed content-months.

3. **Complete Outcome Window:** The following month (t) must contain valid organic session records so we can compute the ground-truth change.

After applying these filters, we were left with **97,692** content-month observations across 18,347 unique content assets—a solid, well-defined modeling population.

2.4 Public Safety and Anonymization Protocols

Data privacy was taken seriously throughout this work:

- Every domain name, hostname, and full page URL was cryptographically hashed before we ever touched the data (fields like `content_hash_id` and `client_hash_id`).
- No raw customer identifiers, proprietary queries, brand keywords, or user IP addresses appear anywhere in this analysis.
- All results are reported at the aggregate statistical level.

3 Methodology

3.1 Label Formulation

We defined the prediction target as a meaningful month-over-month drop in organic traffic. For content entity i at month t , we computed the organic change ratio relative to the prior month:

$$\Delta \text{Organic}_{i,t} = \frac{\text{Organic}_{i,t} - \text{Organic}_{i,t-1}}{\text{Organic}_{i,t-1}} \quad (2)$$

We then set a binary label, $y_{i,t} \in \{0, 1\}$, using a threshold of -20% :

$$y_{i,t} = \begin{cases} 1 & \text{if } \Delta \text{Organic}_{i,t} \leq -0.20 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Across the filtered cohort ($N = 97,692$), **43.82%** of observations ($n = 42,809$) actually experienced this level of decline—confirming that content decay is far from rare, and well worth trying to predict.

3.2 Lagged Feature Engineering

For the model to be useful in practice, every predictor must be available at the end of month $t - 1$, when predictions for month t would be generated. We engineered 23 features ($\mathbf{x}_{i,t} \in \mathbb{R}^{23}$) across four categories:

1. **Search Demand Signals:** Prior-month GSC impressions, organic clicks, CTR, and average SERP position. Figure 2 shows the expected inverse relationship between rank and CTR, consistent with well-known position-bias findings [6].
2. **Multi-Channel Session Attribution:** Session counts broken out by channel—organic, direct, referral, paid, social, and generative AI referrals.
3. **On-Page Engagement:** Pageviews, total users, engaged sessions, engagement rate, average engagement duration, scroll events, and scroll rate.
4. **Momentum Indicators:** Month-over-month growth rates for organic sessions, impressions, clicks, and total traffic (comparing $t - 1$ to $t - 2$).

3.3 Data Leakage Audit

Data leakage is one of the most common and damaging mistakes in applied machine learning—models look great on paper, then collapse the moment you deploy them [7]. If any features computed from month t data accidentally sneak into the training set, performance metrics become meaningless.

To guard against this, we ran an automated assertion check before training:

$$\mathcal{F}_{\text{forbidden}} \cap \mathcal{F}_{\text{features}} = \emptyset \quad (4)$$

where $\mathcal{F}_{\text{forbidden}}$ includes current-period metrics like $\{\text{organic_sessions}_t, \text{clicks}_t, \text{impressions}_t\}$. The audit confirmed that all 23 features come strictly from prior periods—no leakage.

3.4 Time-Aware Validation Design

Standard random cross-validation does not work for forecasting problems. If you randomly shuffle your data, the model ends up “predicting” past events using future information, which defeats the entire purpose [8]. Instead, we used strict chronological splits:

- **Training:** November 2025 – March 2026 ($N_{\text{train}} = 19,621$).

- **Validation:** April 2026 ($N_{\text{val}} = 20,613$), used for tuning and sanity checks.
- **Test (Out-of-Time):** May – June 2026 ($N_{\text{test}} = 57,458$). We did not touch this data until the final evaluation.

3.5 Baseline and Model Architecture

We compared our model against a simple but relevant baseline:

- **No-Decline Baseline:** Always predicts $\hat{y} = 0$ —in other words, assume nothing is declining. This is essentially what happens when an organization takes no proactive action at all.

For the actual model, we built a straightforward, interpretable pipeline:

1. **Median Imputation:** Fills in missing values (e.g., zero denominators in rate calculations) with the median.
2. **Z-Score Standardization:** Centers and scales every feature: $z = (x - \mu)/\sigma$.
3. **Balanced Logistic Regression:** Uses inverse class-frequency weights so the model pays more attention to the minority class and does not just default to predicting “no decline” for everything [9]:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (5)$$

4 Experimental Results

4.1 Comparative Performance on Holdout Test Set

Table 1 summarizes how each approach performed on the held-out test data ($N = 57,458$).

Table 1: Model performance on the chronological test split.

Model / Split	Precision	Recall	F1	ROC-AUC
No-Decline (Test)	0.0000	0.0000	0.0000	—
Logistic Reg. (Val)	0.3769	0.6458	0.4760	0.5348
Logistic Reg. (Test)	0.4967	0.2927	0.3684	0.5409

The baseline, unsurprisingly, catches nothing—it produces zero true positives, so every metric sits at 0. Our logistic regression model, on the other hand, achieves **49.67%** precision and **29.27%** recall on the test set, with an F1 of **0.3684** and an ROC-AUC of **0.5409** (Average Precision = **0.478**).

An ROC-AUC of 0.5409 is admittedly modest when you look at the full probability range. But in practice, what

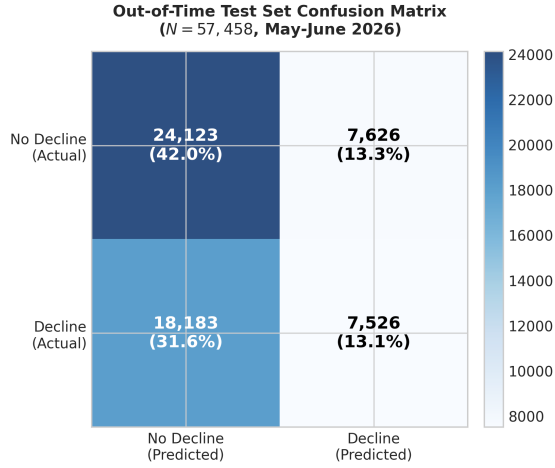


Figure 3: Confusion matrix on the out-of-time holdout set (May–June 2026, $N = 57,458$).

matters most is how well the model performs at the top of its ranked list—the pages it is most confident about. In that decision-critical zone, precision-recall metrics give a much more honest picture than the ROC curve [10].

Table 2: Confusion matrix on the out-of-time test set ($N = 57,458$).

	Pred: No Decline	Pred: Decline
Actual: No Decline	24,123 (TN)	7,626 (FP)
Actual: Decline	18,183 (FN)	7,526 (TP)

4.2 Error Analysis and Confusion Matrix

Table 2 and Figure 3 break down the test predictions. Of the 15,152 pages the model flagged as declining, **7,526 actually did decline**—giving us roughly 50% precision. For a content team managing thousands of pages, that means about half of the pages they review based on the model’s recommendations genuinely needed attention. That is a meaningful improvement over reviewing pages at random.

4.3 Signal Interpretation and Feature Importance

Because we standardized all inputs before fitting, the logistic regression coefficients directly tell us how strongly each feature pushes the prediction toward or away from decline—no need for black-box explanation tricks. Figure 5 shows the most influential features.

Here is what the coefficients tell us:

- **Protective signals** ($\beta < 0$): Prior pageviews ($\beta = -1.7083$), prior organic clicks ($\beta = -0.9252$), and engagement duration ($\beta = -0.2738$) are all associated

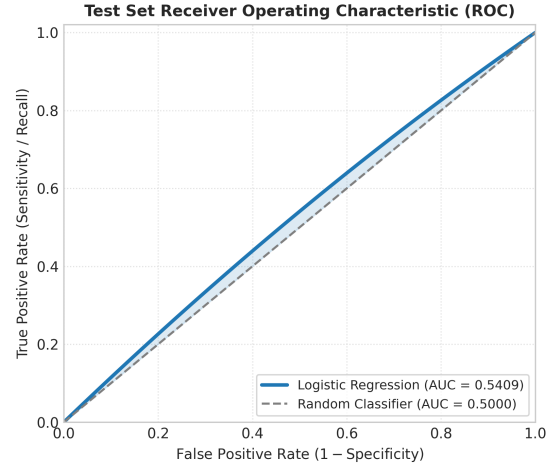


Figure 4: ROC curve on the holdout test set ($AUC = 0.5409$).

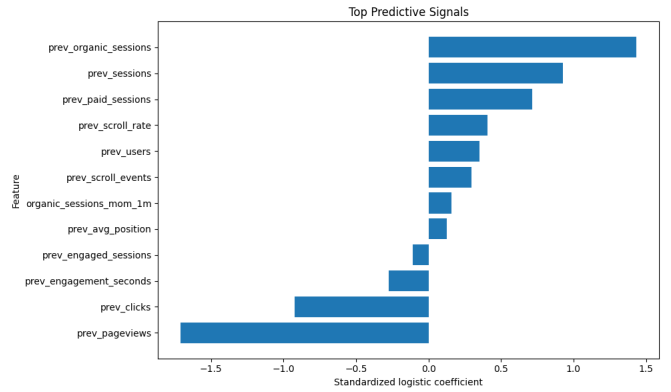


Figure 5: Standardized logistic regression coefficients for the top search and engagement features.

with stability. In plain terms, pages that attract genuine clicks and keep users engaged tend to hold their rankings.

- **Vulnerability signals** ($\beta > 0$): High prior organic sessions ($\beta = +1.4326$), paid sessions ($\beta = +0.7137$), scroll rate ($\beta = +0.4076$), and poor average SERP position ($\beta = +0.1255$) point toward higher decline risk. Pages that had a recent traffic spike or depend heavily on paid promotion without strong organic fundamentals tend to revert back down. And pages sitting on page two or beyond (positions 11–30) are always at risk of getting pushed further out.

4.4 Sensitivity Analysis Across Decline Thresholds

To make sure our results are not just an artifact of the specific -20% cutoff we chose, we re-trained and evaluated the model at -10% and -30% thresholds as well.

Table 3: How model performance varies across different decline definitions.

Threshold	Precision	Recall	F1	ROC-AUC
−10% Decline	0.5687	0.2987	0.3917	0.5574
−20% (Primary)	0.4967	0.2927	0.3684	0.5409
−30% Decline	0.4306	0.2813	0.3403	0.5283

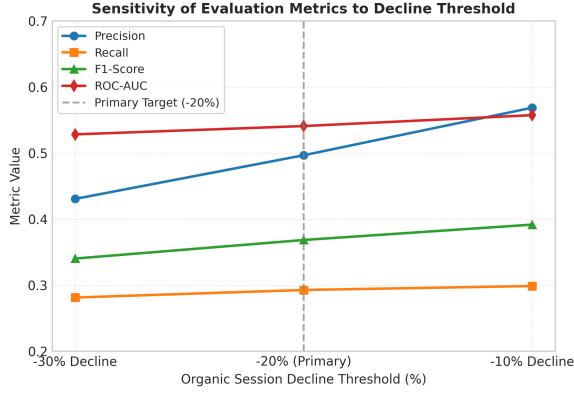


Figure 6: How test-set metrics change across different decline thresholds (−10%, −20%, −30%).

Table 3 and Figure 6 show that the model behaves consistently across all three thresholds. With the more lenient −10% definition, precision climbs to **56.87%** and ROC-AUC reaches **0.5574**. At the stricter −30% level, precision settles at **43.06%**. The fact that performance degrades smoothly rather than erratically tells us the model is picking up on real decay patterns, not just quirks of a particular cutoff value.

5 Ranked Recommendations and Action Playbook

5.1 Decision Engine Architecture

The model outputs a probability for each content entity, but probabilities alone are not actionable. We translated them into a three-tier decision framework. For each item i in the test set, we mapped the predicted probability $p_i = P(y = 1 | \mathbf{x}_i)$ into an action tier:

- **High Priority – Immediate Review** ($p_i \geq 0.75$): These pages face serious decline risk. They go straight to the top of the editorial queue.
- **Medium Priority – Scheduled Monitor** ($0.50 \leq p_i < 0.75$): Moderate risk. Keep a close eye on keyword rankings and check for technical issues on a weekly basis.
- **Low Priority – Protect** ($p_i < 0.50$): Performing well or growing. Leave them alone—unnecessary

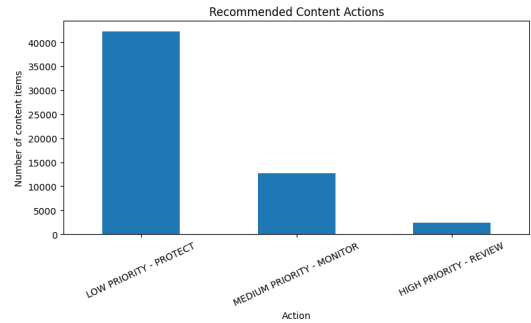


Figure 7: How the 57,458 test-set pages break down across the three action tiers.

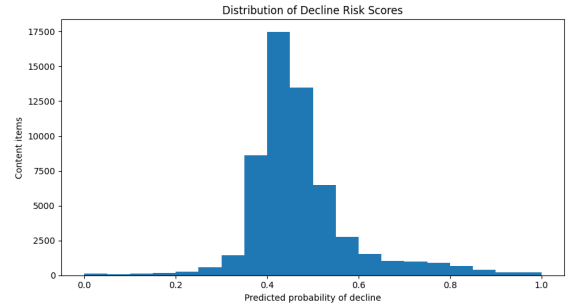


Figure 8: Distribution of predicted decline probabilities across test content items.

changes can do more harm than good.

5.2 Cohort Action Distribution

Across the 57,458 test items (see Figures 7 and 8):

- **High Priority (Review):** 2,384 pages (4.15%). This is a manageable, focused queue—exactly the kind of targeted list a content team can actually work through.
- **Medium Priority (Monitor):** 12,768 pages (22.22%).
- **Low Priority (Protect):** 42,306 pages (73.63%).

5.3 Auditable Editorial Reason Codes

A prediction without an explanation is hard to act on. To make the model’s output genuinely useful for editors, we attach human-readable reason codes to every recommendation:

1. **Low CTR (CTR < 2.0%):** *What to do:* Rewrite title tags and meta descriptions; add schema markup to improve how the page appears in search results.
2. **Weak Average Position (Avg Position > 10):** *What to do:* Expand the content, update outdated

statistics, and build internal links to strengthen relevance.

3. **High Search Visibility (Impressions $\geq 1,000$):** *What to do:* Flag these as high-value assets that deserve priority editorial attention.
4. **Low Engagement Rate (Engagement Rate $< 50\%$):** *What to do:* Run a UX audit—look at readability, remove intrusive popups, and check page load speed.

Table 4 shows the five highest-priority anonymized pages identified by the action engine.

6 Limitations and Honest Framing

No study is without its blind spots, and we want to be upfront about ours:

- **Correlation, Not Causation:** Our model identifies statistical patterns between past behavior and future traffic drops. It does *not* prove that refreshing a page will fix the problem. To establish that kind of causal link, you would need controlled A/B experiments or synthetic control methods.
- **No Insight into Search Algorithms:** Everything in this analysis comes from publisher-side measurement tools (GSC and GA4). We are not modeling or claiming to model how Google’s ranking algorithm actually works.
- **Survivor Bias:** We deliberately excluded pages with fewer than 100 impressions or 10 sessions to avoid noisy percentage swings. That means our findings do not generalize to brand-new or very low-traffic pages.
- **External Shocks:** Google’s core algorithm updates, seasonal shopping trends, and other external events can cause traffic shifts that 30-day behavioral lags simply cannot capture.

7 Reproducibility

We have structured everything for easy replication and future benchmarking:

- **Repository:** The full codebase—data extraction notebooks, model evaluation pipelines, and figure generation scripts—is hosted on GitHub.
- **Data Pipeline:** Built on DuckDB (v1.1+), querying the Hugging Face Parquet release directly at hf://datasets/FlyRank/internship-warehouse.
- **Artifacts:** All model weights, metric summaries, confusion matrices, and ranked output tables are saved:

- `capstone_model_evaluation_results.csv`
- `capstone_feature_importance.csv`
- `capstone_ranked_recommendations.csv`
- `capstone_sensitivity.csv`

8 Acknowledgments and Data Credit

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Built on the FlyRank ML Internship dataset

<https://flyrank.ai>

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References

- [1] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, UK: Cambridge University Press, 2008.
- [2] K. Radinsky, K. M. Svore, S. T. Dumais, M. Shokouhi, J. Teevan, A. Bocharov, and E. Horvitz, “Behavioral dynamics on the web: Learning to predict the popularity of web pages,” *ACM Transactions on the Web (TWEB)*, vol. 7, no. 3, pp. 1–25, 2013.
- [3] T. Joachims, “Optimizing search engines using click-through data,” in *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2002, pp. 133–142.
- [4] FlyRank Research Team, “FlyRank Search Intelligence Data Warehouse: Multi-Tenant Search Console and Analytics Repository,” *FlyRank Machine Learning Internship Series*, 2025, accessed: 2026-06-30. [Online]. Available: <https://flyrank.ai>
- [5] M. Raasveldt and H. Mühleisen, “DuckDB: An embeddable analytical database,” in *Proceedings of the 2019 International Conference on Management of Data (SIGMOD)*, 2019, pp. 1981–1984.
- [6] N. Craswell, O. Zoeter, M. Taylor, and B. Ramsey, “An experimental comparison of click position-bias models,” in *Proceedings of the 2008 International Conference on Web Search and Data Mining (WSDM)*, 2008, pp. 87–94.

Table 4: Top ranked high-priority content recommendations from the out-of-time test cohort.

Rank	Anonymized ID	Month	Prev Org	Prev Imp	Prev CTR	Decline Prob	Reason Codes and Remediation
1	content_fc0e1125...	2026-06	12	4,499	0.27%	1.0000	Low CTR; High Visibility; Low Engagement
2	content_d8a9f276...	2026-05	97	1,324	1.06%	1.0000	Low CTR; Weak Position; High Visibility
3	content_d8a9f276...	2026-06	148	1,908	1.42%	1.0000	Low CTR; Weak Position; High Visibility
4	content_95400d53...	2026-06	2	2,305	0.17%	1.0000	Low CTR; High Visibility; Low Engagement
5	content_5fa2737c...	2026-05	1,862	107,813	1.27%	1.0000	High-Traffic Asset; Low CTR; High Visibility

- [7] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data (TKDD)*, vol. 6, no. 4, pp. 1–21, 2012.
- [8] V. Cerqueira, L. Torgo, and I. Mozetič, “Evaluating time series forecasting models: An empirical study on performance estimation methods,” *Machine Learning*, vol. 109, no. 11, pp. 1997–2028, 2020.
- [9] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [10] T. Saito and M. Rehmsmeier, “The precision-recall plot is more informative than the ROC plot when evaluating imbalanced datasets,” *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.